COS 201

COMPUTER PROGRAMMING I

Assignments and Practical Exercises — Weeks 1–3

Student Name: NNADI GABRIEL UGOCHUKWU

Matriculation Number: 25/18103

Course: COS 201 — Computer Programming I

Programme: Software Engineering

GitHub Repository: https://github.com/nnadigabriel/COS-201

Week 1 — Practical: FirstProgram.java

This program displays the course information, programme of study, three reasons programming matters to the discipline, and a learning goal.

```
public class FirstProgram {
    public static void main(String[] args) {
        System.out.println("Course: COS 201 - Computer Programming I");
        System.out.println("Programme: Software Engineering");
        System.out.println("Reason 1: Programming helps solve computational problems.");
        System.out.println("Reason 2: Programming supports automation of tasks.");
        System.out.println("Reason 3: Programming enables software development.");
        System.out.println("My goal: I want to develop simple working applications.");
    }
}
```

Expected output

```
Course: COS 201 - Computer Programming I
Programme: Software Engineering
Reason 1: Programming helps solve computational problems.
Reason 2: Programming supports automation of tasks.
Reason 3: Programming enables software development.
My goal: I want to develop simple working applications.
```

Week 2 — Practical: HelloWorld.java

A basic Java program that prints a greeting and demonstrates the class, main method and output statement.

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, COS 201!");
        System.out.println("My name is NNADI GABRIEL UGOCHUKWU.");
        System.out.println("Matriculation Number: 25/18103");
    }
}
```

Expected output

```
Hello, COS 201!
My name is NNADI GABRIEL UGOCHUKWU.
Matriculation Number: 25/18103
```

Week 2 — Student Profile Practical

```
import java.util.Scanner;

public class StudentProfile {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = input.nextLine();

        System.out.print("Enter your programme: ");
        String programme = input.nextLine();

        System.out.print("Enter your level: ");
        int level = input.nextInt();

        System.out.println("\n--- STUDENT PROFILE ---");
        System.out.println("Name: " + name);
        System.out.println("Programme: " + programme);
        System.out.println("Level: " + level);
        input.close();
    }
}
```

Week 3 — Practical 1: SimpleCalculator.java

Reads two numbers and displays their sum, difference, product and quotient. Division by zero is handled explicitly.

```
import java.util.Scanner;

public class SimpleCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the first number: ");
        double firstNumber = input.nextDouble();

        System.out.print("Enter the second number: ");
        double secondNumber = input.nextDouble();

        System.out.printf("Sum: %.2f%n", firstNumber + secondNumber);
        System.out.printf("Difference: %.2f%n", firstNumber - secondNumber);
        System.out.printf("Product: %.2f%n", firstNumber * secondNumber);

        if (secondNumber != 0) {
            System.out.printf("Quotient: %.2f%n",
                              firstNumber / secondNumber);
        } else {
            System.out.println("Division is undefined for a zero divisor.");
        }
        input.close();
    }
}
```

Sample run

```
Enter the first number: 20
Enter the second number: 4
Sum: 24.00
Difference: 16.00
Product: 80.00
Quotient: 5.00
```

Week 3 — Practical 2: AverageCalculator.java

```
import java.util.Scanner;

public class AverageCalculator {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = input.nextLine();

        System.out.print("Enter score for Course 1: ");
        double score1 = input.nextDouble();
        System.out.print("Enter score for Course 2: ");
        double score2 = input.nextDouble();
        System.out.print("Enter score for Course 3: ");
        double score3 = input.nextDouble();

        double total = score1 + score2 + score3;
        double average = total / 3.0;

        System.out.println("\n--- STUDENT RESULT ---");
        System.out.println("Student: " + name);
        System.out.printf("Total: %.2f%n", total);
        System.out.printf("Average: %.2f%n", average);
        input.close();
    }
}
```

Sample run

```
Enter student name: NNADI GABRIEL UGOCHUKWU
Enter score for Course 1: 70
```

```
Enter score for Course 2: 80
Enter score for Course 3: 90

--- STUDENT RESULT ---
Student: NNADI GABRIEL UGOCHUKWU
Total: 240.00
Average: 80.00
```

Week 3 — Practical 3: ItemCost.java

```
import java.util.Scanner;

public class ItemCost {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter item name: ");
        String itemName = input.nextLine();

        System.out.print("Enter quantity: ");
        int quantity = input.nextInt();

        System.out.print("Enter unit price: ");
        double unitPrice = input.nextDouble();

        double totalCost = quantity * unitPrice;

        System.out.println("\n--- PURCHASE SUMMARY ---");
        System.out.println("Item: " + itemName);
        System.out.println("Quantity: " + quantity);
        System.out.printf("Unit price: %.2f%n", unitPrice);
        System.out.printf("Total cost: %.2f%n", totalCost);
        input.close();
    }
}
```

Sample run

```
Enter item name: Exercise book
Enter quantity: 3
Enter unit price: 1500
--- PURCHASE SUMMARY ---
Item: Exercise book
Quantity: 3
Unit price: 1500.00
Total cost: 4500.00
```

Week 3 — Debugging Exercise

Original expression: `double result = firstNumber / secondNumber;` When both operands are int, Java performs integer division before converting the result to double. Cast one operand to double to obtain a fractional result.

```
int firstNumber = 5;
int secondNumber = 2;
double result = (double) firstNumber / secondNumber;
System.out.println(result); // 2.5
```

Practical Submission Message — Week 2

```
COS 201 — Week 2 Practical Submission

Dear Lecturer,

I am submitting my Week 2 practical assignment for
COS 201 — Computer Programming I.

Student Name: NNADI GABRIEL UGOCHUKWU
Matriculation Number: 25/18103
```

Programme: Software Engineering
Assignment: Development Environment and First Java Program
GitHub Repository: https://github.com/nnadigabriel/COS-201

I have included my Java source code and README documentation
in the repository. I have compiled and executed the program
to verify the output.

Thank you.